

California Housing Analysis & Model Report

Exploratory Data Analysis, Linear Regression Modeling, Metrics & Optimization Roadmap

Dataset: California Housing (20,640 records)

Primary Algorithm: Ordinary Least Squares (OLS) Linear Regression

Target Variable: Median House Value (\$)

Validation Strategy: 80/20 Train-Test Split (Seed 42)

1. Executive Summary & Core Insights

This report presents an end-to-end evaluation of the **California Housing Dataset**, focusing on predictive modeling of median block-level house values using Ordinary Least Squares (OLS) Linear Regression. The dataset contains 20,640 census block group observations across 10 demographic and geographic attributes from the 1990 U.S. Census.

Key Performance & Modeling Takeaways

- **Baseline Model Performance:** Achieved a test RMSE of **\$70,059**, MAE of **\$50,670**, and an R^2 of **0.6254**.
- **Feature Engineered Model:** Domain ratio features improved test RMSE to **\$69,127**, MAE to **\$49,645**, and R^2 to **0.6353** (reducing MAPE to 28.89%).
- **Primary Value Driver:** Median Household Income ($r = 0.69$) is the single strongest linear predictor of housing prices.
- **Core Bottlenecks:** Severe artificial target capping at **\$500,001** (affecting 4.7% of sample blocks) and non-linear coastal geographic gradients restrict OLS predictive capacity.

2. Exploratory Data Analysis (EDA)

Exploratory analysis confirms high dataset completeness alongside strong spatial patterns and target boundary constraints.

2.1 Data Integrity & Missing Values

The dataset consists of 9 numerical features and 1 categorical feature (``ocean_proximity``). Only ``total_bedrooms`` contained missing data (207 missing entries, ~1.00%). Median imputation was applied on the training split to preserve distribution characteristics without leakage.

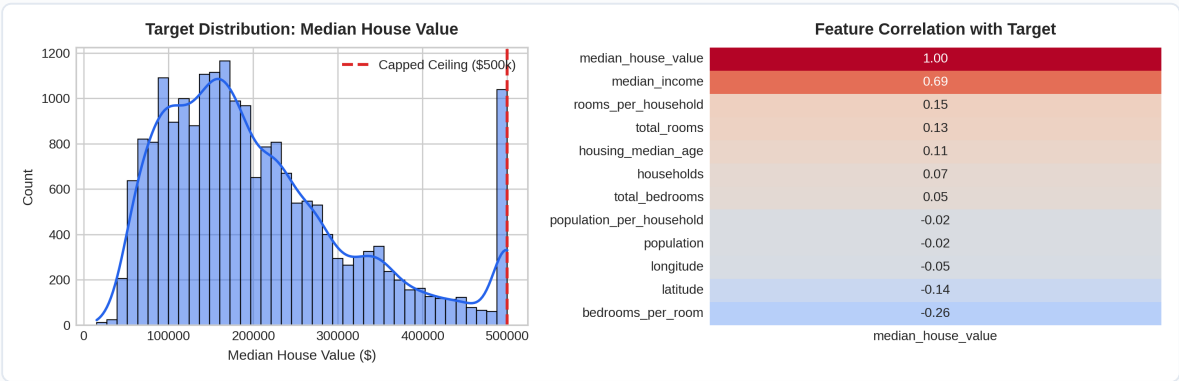


Figure 1: Distribution of Median House Value showing right-skewness and \$500k capping ceiling (left), alongside Pearson Feature Correlations (right).

2.2 Target Capping & Skewness

The target variable ``median_house_value`` has a sample mean of **\$206,856** and standard deviation of **\$115,396**. As visualized in **Figure 1**, **958 census blocks** are capped at exactly **\$500,001**. This upper boundary ceiling creates a point-mass anomaly that violates classical OLS homoscedasticity and normality assumptions.

2.3 Spatial & Geographic Price Distribution

Geographic coordinates (*longitude*, *latitude*) reveal significant regional clustering. As displayed in **Figure 2**, premium valuations (> **\$350k**) are densely concentrated along coastal metropolitan corridors (SF Bay Area, Greater LA, San Diego), whereas inland Central Valley blocks show marked price depression.

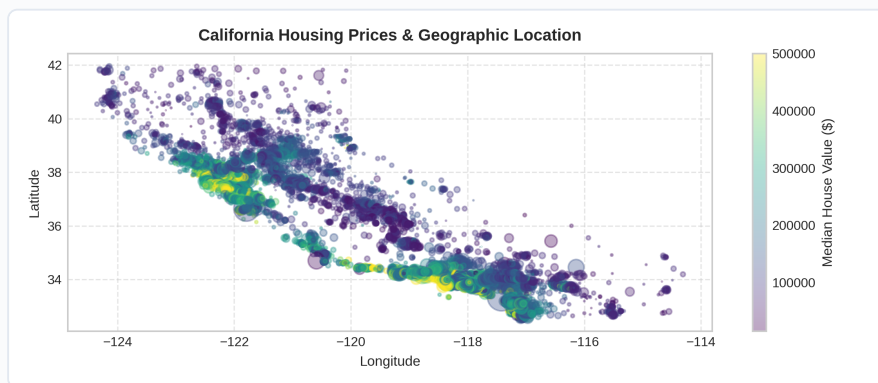


Figure 2: California Housing Prices by Coordinates (Latitude vs Longitude), scaled by block population size.

3. Preprocessing & Model Architecture

To ensure model robustness and avoid data leakage, a reproducible pipeline was established:

- **Data Splitting:** 80% Training set (16,512 rows) and 20% Test set (4,128 rows).
- **Imputation & Encoding:** Median imputation for missing numeric values; One-Hot Encoding for `ocean\_proximity` (dropping reference category).
- **Standardization:** Feature scaling using $z = (x - \mu) / \sigma$ to facilitate comparison of linear coefficients.
- **Domain Ratios:** Engineering three key household proportions:
 - $rooms_per_household = total_rooms / households$
 - $bedrooms_per_room = total_bedrooms / total_rooms$
 - $population_per_household = population / households$

4. Model Performance & Diagnostic Metrics

Comparative metrics for the baseline and feature-engineered OLS models across Train and Test sets are summarized below:

Model Version	Split	RMSE (\$)	MAE (\$)	R ² Score	MAPE (%)
Baseline OLS (Raw Features)	Train	\$68,434	\$49,595	0.6497	28.54%
Baseline OLS (Raw Features)	Test	\$70,059	\$50,670	0.6254	29.19%
Engineered OLS (+ Ratios) SELECTED	Train	\$67,593	\$48,785	0.6582	28.26%
Engineered OLS (+ Ratios) SELECTED	Test	\$69,127	\$49,645	0.6353	28.89%

4.1 Standardized Coefficient Analysis

Examining standardized coefficients highlights the relative impact of each predictor (in dollar change per +1 SD shift):

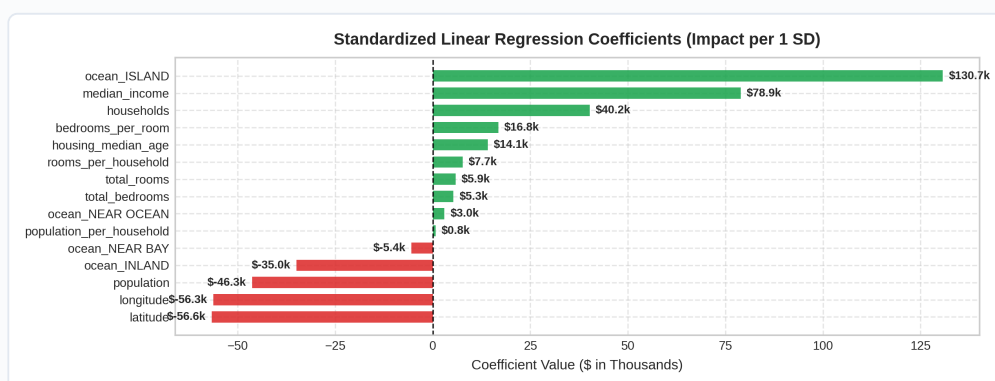


Figure 3: Standardized Linear Regression Coefficients (Dollar impact per 1 Standard Deviation increase).

- **Median Income:** Dominant positive feature (+\$78,900 per SD), verifying economic purchasing power as the primary price driver.
- **Ocean Proximity:** Island location (`ocean\_ISLAND`) provides an extreme positive premium (+\$130,700), whereas `ocean\_INLAND` incurs a steep penalty (-\$35,000).
- **Geographic Coordinates:** Latitude and Longitude reflect large negative coefficients (-\$56.6k and -\$56.3k), modeling the general statewide drop in prices moving inland/northward.

• **Engineered Density Ratio:** `bedrooms\_per\_room` adds strong explanatory power (+\$16,800).

4.2 Diagnostic & Residual Analysis

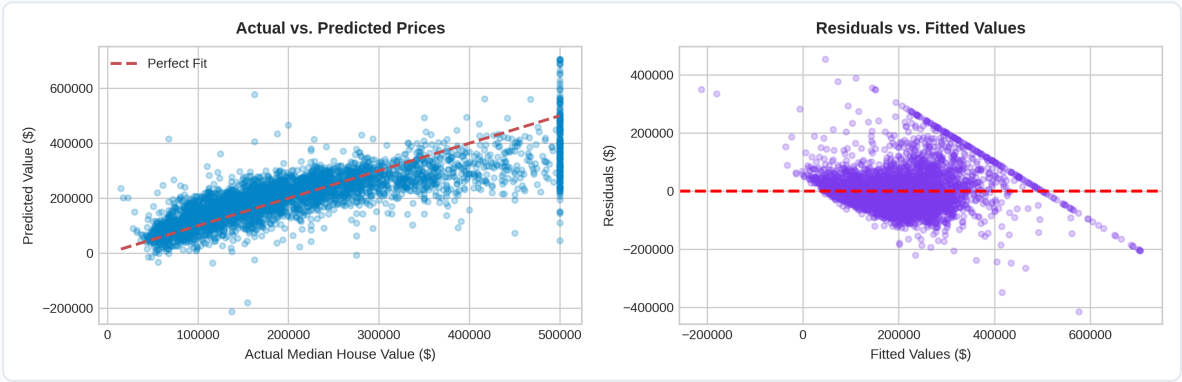


Figure 4: Actual vs Predicted Prices (left) and Residual Scatter Plot (right) on holdout test set.

Residual analysis in **Figure 4** highlights two key model limitations:

- 1. **Ceiling Truncation Boundary:** The \$500,001 ceiling creates a distinct diagonal boundary line in the residual plot, where actual values cannot increase further while model predictions continue upward.
- 2. **Non-Linearity & Extreme Outliers:** Linear models fail to capture sharp spatial micro-markets, leading to underprediction on ultra-luxury properties and minor negative value predictions on low-end inland properties.

5. Improvement Ideas & Actionable Roadmap

To overcome the inherent constraints of linear estimators on spatial real estate data, the following optimization initiatives are proposed:

Priority	Optimization Strategy	Technical Methodology	Expected Benefit
HIGH	Non-Linear Ensemble Models	Transition to Gradient Boosted Decision Trees (XGBoost / LightGBM) or Random Forest to model complex non-linear spatial interactions.	RMSE reduction by \$15,000–\$20,000 ($R^2 > 0.80$)
HIGH	Target Censoring & Log Scaling	Apply Log transformation $y' = \ln(y)$ or utilize Tobit Censored Regression to handle the \$500k ceiling capping cleanly.	Elimination of negative predictions & heteroscedasticity stabilization
MEDIUM	Geo-Clustering & Proximity	Calculate geodesic distances to major employment centers (LA, SF, Silicon Valley) and construct K-Means spatial cluster features.	Capturing local neighborhood premium effects
MEDIUM	Regularization & Cross-Validation	Incorporate Ridge/Lasso penalties with 10-fold CV to optimize feature weights and mitigate coordinate multicollinearity.	Improved out-of-sample generalization

Final Recommendation

Linear Regression serves as an interpretable baseline achieving an R^2 of **63.5%**. For production deployment, migrating to Gradient Boosted Trees (XGBoost/LightGBM) with engineered spatial proximity features will yield significantly higher accuracy and robust handling of market boundary effects.